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Motivation Letter 7. Oktober 2024

To whom it may concern,

I am writing to apply for the PhD posotion focusing on the Big Model in the Artificial Intelligence at your university.

With a deep-seated passion for Computer Science and Mathematics, I began my studies in Computational Mathematics in 2019 after earning my Bachelor's degree in Information Security in China. While focusing on Deep Learning technology, I participated in a research project aimed at imitating personal handwriting styles. The project involved learning the Chinese handwritten style of Shiing Shen Chern from approximately 220 characters. Through this experience, I gained knowledge of Generative Adversarial Networks and developed skills in manuscript searching, data preprocessing, and code replication. As a result, I completed my Master's thesis on the personalized imitation of Chinese handwritten characters using GANs.

Furthermore, to integrate Deep Learning with numerical approximation theory, I participated in the Compressed Sensing MRI ADMM-Net project. Traditional numerical algorithms perform well on simple images, but they fail to preserve detailed information after numerous iterations. With advancements in Deep Learning, many researchers have employed Convolutional Neural Networks to unfold equations and automatically adjust parameters during iterations to preserve details. Through this project, I gained knowledge of convergent algorithm theory and proofs, as well as the integration process with Deep Learning. This experience significantly enhanced my skills in both programming and theoretical understanding.

After earning my Master's Degree in Computational Mathematics, I worked as a Research Assistant in a group at Sun Yat-sen University dedicated to Computer-Assisted Intervention. During my tenure, I participated in proposal writing for a key program from the Science and Technology of China (2023YFE0204300, which was successfully approved). I also completed two progress reports and the completion report for projects funded by the National Natural Science Foundation of China (Grants 81971691 and 12126610). Additionally, I wrote the technological specifications for a patent application and a medical device application. Over the past two years, I have contributed to three projects focused on diagnosing Placenta Accreta Spectrum Disorders, predicting metastasis in Axillary Sentinel Lymph Nodes, and predicting the response to Neoadjuvant Chemotherapy based on MRI. The first two projects yielded excellent experimental results but lacked technological novelty. Throughout these tasks, I consolidated skills in programming (Python, PyTorch, and TensorFlow) and manuscript reading.

I believe that the PhD position in your research unit will be a life-changing experience to achieve the career I aspire for. Your research field the Machine Learning Mechanism are aligned my core interests. I am genuinely committed to investigating pragmatic ways to maximize use of Engineering and Computer Science.

I am confident that I can thrive as a promising candidate for this PhD position. My academic background and program experience, along with my personal skills, demonstrate my compatibility with this program. In summary, I have acquired extensive knowledge and skills in Computer Science and Computational Mathematics. Additionally, my research in Deep Learning, Medical Science, and Mathematics has equipped me to be a competent team player.

Until then, I am grateful for the opportunity to apply to this institution and I am ready to discuss my future with you should my candidacy prove successful.

Sincerely,

Hai Jiang

Attached: curriculum vitæ

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EDUCATIONAL

2019–2022 **Master of Science in Computational Mathematics**, ***Nankai University***, **Tianjin**, **China**

Thesis *GANs based Personal style imitation of Chinese handwritten characters*

Advisors Prof. Yunhua Xue, Prof. Chunlin Wu

Related Courses Approximation theory and methods, Numerical Optimization, Convex Analysis, Variational Analysis, Real Analysis, Functional Analysis, Matrix Computation, Measure Theory and Probability Theory Basis, Numerical Solution of Partial Differential Equations, etc.

2014–2018 **Bachelor of Engineering in Information Security**, ***Lanzhou University***, **Lanzhou**, **China**

Thesis *Improved Upper Bounds of Roman Domination Number in Maximal Outerplanar Graphs*

Advisor Prof. Zepeng Li

Related Courses Discrete Mathematics, Operating System, Data Structure, C and C++ Programming Laboratory, Java Programming Laboratory, Database Theory and Laboratory, Computer Organization and Design, etc

RESEARCH EXPERIENCE

Sun Yat-sen University, School of Computer Science and Engineering

11.2022–7.2024 **Computational Medical Image Laboratory**, Research Assistant

- **Project:** China Department of Science and Technology Key Grant, focused on Breast Cancer, aims to develop models with clinical interpretability and generalization.
- **Supervisors:** Prof. Yao Lu, Dr. Ting Song
- **Focus:** Placenta Accreta Spectrum Disorders, T2-WI MRI, Prognosis, Multi-class classification.
- **Experience and Skills:** Literature research, data preprocessing, model building(programming).
- **Publication:** Preparing.

12.2023–1.2024 **Computational Medical Image Laboratory**, Research Assistant

- **Project:** National Natural Science Foundation of China, focused on Breast Cancer, aimed to develop a prediction model for the Chinese female population mainly with FFDM and US
- **Supervisors:** Prof. Yao Lu, Dr. Xiang Zhang
- **Focus:** Breast Cancer, Dual-Energy CT, Sentinel Lymph Nodes, Metestatic status, Multi-class classification.
- **Experience and Skills:** The first complete research experience included conducting literature research, experiment design, paper writing, and Tensorflow framework.
- **Publication:** Submitted to MICCAI2024; in Revising.

Nankai University, School of Mathematical Sciences

1.2022–6.2022 **Research Student**, Student

- **Project:** ADMM model from the manuscript ”Deep ADMM-Net for Compressed-Sensing MRI.”
- **Supervisors:** Prof. Chunlin Wu, Prof. Yunhua Xue
- **Focus:** Compressed-sensing Theory, Iterative Equations, Neural Networks, MRI reconstruction.
- **Experience and Skills:** The second programming experience included proof of mathematical equations and Deep Learning. I have reproduced the mathematical iteration equations in C++, Python, and PyTorch.

1.2021–4.2021 **Research Student**, Student

■ **Project:** ROF-model from the manuscript "Nonlinear Total Variation Based Noise Removal Algorithms."

■ **Supervisors:** Prof. Yunhua Xue

■ **Focus:** Image Restoration, Denoise, PDE, Total-Variation Penalty.

■ **Experience and Skills:** Numerical experiments used both C++ and Python to build the ROF model.

OTHER WORK EXPERIENCE

Funding

Proposal Proposal Writing; Succeeded; China Department of Science and Technology Key Grant 2023YFE0204300.

Report Finished 2 Completion Report and 2 Progress Report; Succeeded; NSFC Grant 81971697, 12126610.

Specification

Patent 1 Application Specification; in Progress.

Device 1 Medical Device Specification; in Progress.

Teaching Assistant

Undergraduate Courses Calculus; Mathematical Analysis

LANGUAGE PROFICIENCY

Mandarin

Native Speaker

English

IELTS 6.5; CET6 476/710; CET4 544/710; Fluent(speaking, reading, writing).

Cantonese

Intermediate

SKILLS

Professional

Python, PyTorch, Tensorflow, C/C++, LaTeX, MATLAB, Mathematics

Other

Linux, Microsoft Office, Adobe Photoshop

INTEREST

Artificial Intelligence; Mathematics; Physics

REFERENCES

Prof. Yao Lu

Professor of Computational Mathematics and Medical Image Analysis at Sun Yat-sen University

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